

$$\sin(\pi x) \quad x \in [-\frac{1}{4}; -\frac{1}{6} \cup 0; \frac{1}{4}]$$

$$i=0, 1, 2, 3 \quad n=3$$

1. Построение интерполяционного полинома в до-ме Ньютона

$$x_0 = -\frac{1}{4}, f(x_0) = \sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$$

$$x_1 = -\frac{1}{6}, f(x_1) = \sin(-\frac{\pi}{6}) = -\frac{1}{2}$$

$$x_2 = 0, f(x_2) = \sin(0) = 0$$

$$x_3 = \frac{1}{4}, f(x_3) = \sin(\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$$

$$f(x_0, x_1) = \frac{-\frac{1}{2} + \frac{\sqrt{2}}{2}}{-\frac{1}{6} + \frac{1}{4}} = \frac{\sqrt{2} - 1}{\frac{1}{6}} = 6\sqrt{2} - 6$$

$$f(x_1, x_2) = \frac{0 + \frac{1}{2}}{0 + \frac{1}{6}} = 3$$

$$f(x_2, x_3) = \frac{\frac{\sqrt{2}}{2} - 0}{\frac{1}{4} - 0} = 2\sqrt{2}$$

$$f(x_0, x_1, x_2) = \frac{-(6\sqrt{2} - 6) + 3}{0 + \frac{1}{4}} = \frac{3 - (6\sqrt{2} - 6)}{\frac{1}{4}} = \frac{3 + 6 - 6\sqrt{2}}{\frac{1}{4}} = 36 - 24\sqrt{2}$$

$$f(x_1, x_2, x_3) = \frac{2\sqrt{2} - 3}{\frac{1}{4} + \frac{1}{6}} = \frac{(2\sqrt{2} - 3)12}{5} = \frac{24\sqrt{2} - 36}{5}$$

$$f(x_0, x_1, x_2, x_3) = \frac{\frac{24\sqrt{2} - 36}{5} - (36 - 24\sqrt{2})}{\frac{1}{4} + \frac{1}{4}} = \frac{288\sqrt{2} - 432}{5}$$

$$N_3(x) = f(x_0) + f(x_0, x_1)(x - x_0) + f(x_0, x_1, x_2)(x - x_0)(x - x_1) + f(x_0, x_1, x_2, x_3)(x - x_0)(x - x_1)(x - x_2) =$$

$$= -\frac{\sqrt{2}}{2} + (6\sqrt{2} - 6)(x + \frac{1}{4}) + (36 - 24\sqrt{2})(x + \frac{1}{4})(x + \frac{1}{6}) + \frac{(288\sqrt{2} - 432)}{5}(x + \frac{1}{6})(x + \frac{1}{5})x = \frac{88\sqrt{2} - 432}{5}x^2 + \frac{288\sqrt{2} - 432}{5}x - \frac{\sqrt{2}}{2}$$

2.3. Вычисление оценки остаточного полинома. Сравнение

с значением интерполяционного полинома и реальной переменной. где точек $-\frac{1}{5}$ и $\frac{1}{5}$.

$$a) x^0 = -\frac{1}{5}$$

$$\omega(-\frac{1}{5}) = (-\frac{1}{4} + \frac{1}{5})(-\frac{1}{6} + \frac{1}{5})(\frac{1}{5})(\frac{1}{4} + \frac{1}{5}) = -\frac{3}{20000}$$

$$f(x) = \sin(\pi x), x \in [-\frac{1}{4}; \frac{1}{4}]$$

$$M_4 = \max |f^{(4)}(x)| \Rightarrow M_4 = \frac{\pi^4 \sqrt{2}}{2}; f^{(4)}(x) = \pi^4 \sin(\pi x)$$

$$\frac{\pi^4 \sqrt{2}}{2} \cdot \frac{1}{4!} \cdot \frac{3}{20000} \approx 0,00086$$

$$|f(x^0) - N_3(x^0)| \geq |\sin(-\frac{\pi}{5}) + \frac{88\sqrt{2} - 432}{625}| \approx 0,786 \geq \frac{\pi^4 \sqrt{2}}{2 \cdot 4!} \cdot \frac{3}{20000}$$

$$\delta) x^0 = \frac{1}{5}$$

$$W\left(\frac{1}{5}\right) = \frac{-33}{20000}$$

$$\mu_4 = \frac{\pi^4 \sqrt{2}}{2} \Rightarrow \frac{\pi^4 \sqrt{2}}{2} \frac{1}{4!} \frac{33}{20000} \approx 0,0017$$

$$|f(x) - N_3(x)| = \left| \sin\left(\frac{\pi}{5}\right) - \frac{88\sqrt{2} + 243}{625} \right| \approx 0,00014 < \frac{\pi^4 \sqrt{2}}{2} \frac{1}{4!} \frac{33}{20000}$$

~~снова~~

4. График погрешности

$$r(x) = f(x) - N_3(x) = \sin(\pi x) - \frac{x^3(288\sqrt{3} - 432) + x(27 - 8\sqrt{2})}{5}$$

